

BEATMAMBA: BIDIRECTIONAL SELECTIVE STATE-SPACE MODELING FOR EFFICIENT BEAT TRACKING

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ABSTRACT

Accurate beat tracking is fundamental to music analysis, yet remains challenging due to its requirement for simultaneous modeling of local transient onsets and global rhythmic states. While recent state-of-the-art methods leverage Transformers for long-range context modeling, their quadratic complexity hinders efficiency on long music sequences. To address these limitations, we introduce BeatMamba, a novel architecture that unifies local precise positioning and global rhythmic coherence through selective state space model. The model's input-dependent selection dynamically focuses computation on sparse beat events, while its continuous state evolution captures rhythmic continuity and hierarchical dependencies. Additionally, grounded in the temporal regularity of musical rhythm, we propose a rhythmic consistency loss to enforce structural stability and temporal alignment in beat predictions. Comprehensive evaluations on benchmark datasets demonstrate BeatMamba achieves superior performance in beat tracking, highlighting its potential as an efficient and robust solution for complex musical scenarios.

Index Terms— beat tracking, selective state-space model, rhythmic consistency loss, inter-beat interval regularization

1. INTRODUCTION

Beat tracking is the computational task of identifying metrical pulse positions in music audio. It is fundamental to critical applications including music transcription [1, 2] and music structure analysis [3, 4]. Despite decades of research, systems still struggle to achieve human-level robustness, particularly when processing passages featuring weak onsets and polyphonic textures. Fundamentally, these challenges stem from the dual-scale modeling dilemma: local transient detection requires fine-grained resolution, while global rhythmic coherence demands long-range dependency modeling.

Early approaches leveraged Recurrent Neural Networks [5, 6] for temporal modeling but struggled with long-range

dependencies due to vanishing gradients. Convolutional Neural Networks [7, 8] later dominated with local feature extraction prowess, yet their limited receptive fields hindered global rhythmic state tracking. Temporal Convolutional Networks [9, 10, 11] mitigated this via dilated convolutions but incurred exponential depth costs for full-sequence modeling. Recent state-of-the-art methods [12, 13, 14, 15, 16] address this via Transformer architectures, leveraging self-attention to model arbitrary-length dependencies. However, their quadratic computational complexity ($O(N^2)$) in sequence length incurs prohibitive costs for long music recordings, severely limiting practical deployment in real-world scenarios.

To resolve these limitations, we introduce BeatMamba, a novel hybrid architecture that leverages convolutional layers for local feature extraction and state space model (SSM) [17] for global rhythmic reasoning. BeatMamba begins with convolutional layers that perform initial feature projection and reduce the temporal resolution to a computationally tractable scale. The core of BeatMamba is a symmetrical U-shape framework constructed entirely from Mamba blocks. This design enables linear-time ($O(N)$) modeling of long-range dependencies across the entire sequence, allowing the model to capture everything from fine-grained temporal nuances to overarching rhythmic structures. Further, inspired by music's temporal regularity principle, we introduce a rhythmic consistency (RC) loss. By penalizing variance in predicted inter-beat intervals, the RC loss explicitly enforces temporal regularity aligned with musical meter, guiding the model toward structurally coherent and musically plausible predictions.

The main contributions of this paper are as follows: 1) We present the first integration of state space models into beat tracking via BeatMamba, establishing a new paradigm for efficient long-context rhythm analysis. 2) BeatMamba synergistically integrates convolutional layers for local feature extraction with state space models for global rhythmic reasoning, resolving the inherent efficiency issues of Transformer-based methods. 3) We introduce a rhythmic consistency loss that explicitly enforces temporal regularity in the predicted beat sequences, significantly enhancing their structural coherence. 4) Comprehensive evaluations demonstrate BeatMamba achieves superior performance on beat tracking.

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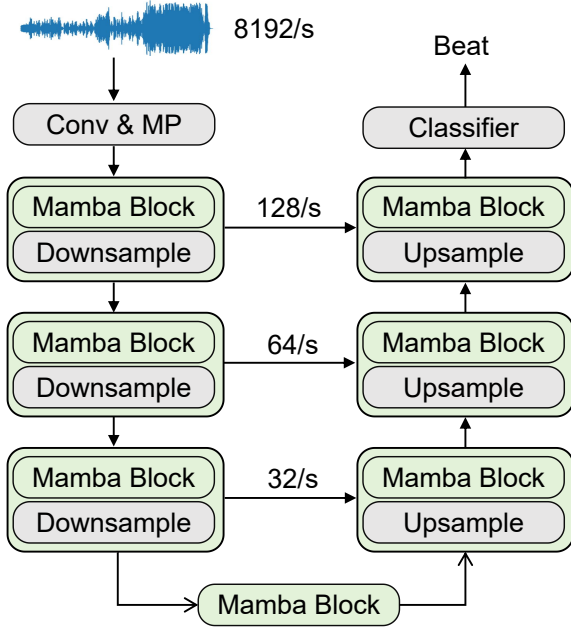


Fig. 1. Overall architecture of the proposed BeatMamba.

2. METHOD

2.1. Network Architecture

The overall architecture of BeatMamba is depicted in Fig. 1. It adopts a hybrid encoder-decoder framework designed for hierarchical processing of audio sequences in beat tracking. This design synergistically integrates convolutional operations for local feature extraction with state-space models for global sequential modeling, thereby simultaneously achieving local precise positioning and long-range rhythmic coherence.

Given an input raw audio waveform $I \in R^{S \times T}$, where S and T representing the sampling rate and duration respectively, BeatMamba first employs a temporal compression encoder. This encoder comprises three sequential 1D convolutional layers each followed by max-pooling (stride=4), transforming the audio into compact temporal features $F \in R^{L \times C}$, where $L = \frac{S}{64} \times T$ and C represent channel dimension.

The core of BeatMamba employs a symmetrical U-shaped encoder-decoder backbone entirely composed of Mamba blocks. The encoder progressively compresses temporal resolution through hierarchical downsampling to extract multi-scale rhythmic representations. The decoder inversely reconstructs the original resolution via mirrored upsampling, integrating skip-connected encoder features to preserve fine-grained temporal information. All Mamba blocks implement temporal bidirectional scanning to holistically model rhythmic dependencies by synthesizing past and future contexts.

Finally, the decoder output is processed by a classifier to generate the final beat activation probability sequence.

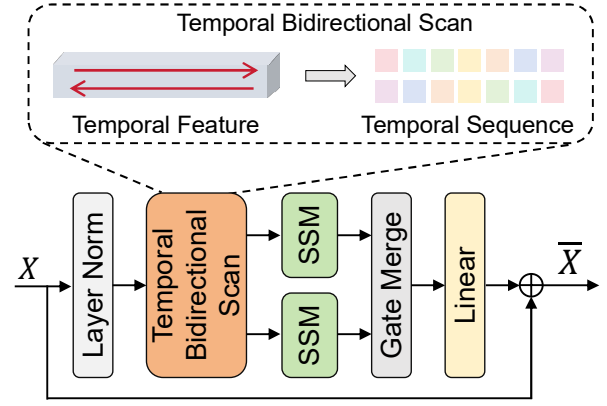


Fig. 2. Detailed architecture of the Temporal Bidirectional Mamba Block.

2.2. Temporal Bidirectional Mamba Block

As illustrated in Fig. 2, the temporal bidirectional Mamba block is the fundamental building block of our architecture, designed to enable long-range, efficient sequence modeling with full contextual awareness.

Given input features $X \in R^{L \times C}$, the block first applies Layer Normalization to stabilize the training process. The normalized features then undergo temporal bidirectional scanning, which processes the sequence in both forward and backward directions simultaneously. This bidirectional operation enables the model to capture contextual information from both past and future time steps, crucial for understanding rhythmic patterns in audio signals. The scanned features are subsequently processed through two parallel state space model (SSM) layers [17] to obtain the forward hidden feature $X_{forw} \in R^{L \times C}$ and backward hidden feature $X_{back} \in R^{L \times C}$:

$$X_{forw} = SSM^{forw}(LN(X)), \quad (1)$$

$$X_{back} = SSM^{back}(LN(X)). \quad (2)$$

These SSM layers employ a selection mechanism that dynamically adjusts their parameters based on the input, allowing the model to focus computational resources on salient rhythmic events while maintaining linear complexity with respect to sequence length. The outputs from both SSM pathways are then combined using a learnable gate merge mechanism. Unlike simple concatenation or summation, this gated fusion dynamically weights the contributions from each pathway, enabling more flexible integration of bidirectional contextual information. Finally, the merged features are projected through a linear layer, and a residual connection adds the original input X to produce the final output $\bar{X} \in R^{L \times C}$:

$$\bar{X} = X + Linear(G(X_{forw}, X_{back})). \quad (3)$$

where $G(\cdot)$ denotes the gate merge operation that adaptively fuses X_{forw} and X_{back} through learnable gating weights.

2.3. Rhythmic Consistency Loss

Existing beat tracking methods typically employ frame-level losses (e.g., Binary Cross-Entropy) that supervise predictions at individual time steps. While effective for local detection, such losses fail to explicitly enforce the temporal regularity inherent to musical rhythm. This often leads to predictions with inconsistent inter-beat intervals, compromising structural coherence.

To guide the model towards generating structurally coherent and musically plausible beat sequences, we introduce a rhythmic consistency (RC) loss. The RC Loss is grounded in the musicological principle of isochrony, defined as the tendency of beats to occur at approximately regular time intervals in most musical pieces. Formally, it assumes that the sequence of Inter-Beat Intervals (IBIs) between consecutive predicted beats should exhibit low variability under stable tempo conditions. Given the predicted beat times $\{\hat{t}_1, \hat{t}_2, \dots, \hat{t}_N\}$, we define the IBIs as:

$$\Delta_i = \hat{t}_{i+1} - \hat{t}_i, \quad i = 1, \dots, N-1. \quad (4)$$

To improve stability and avoid the dominance of large raw variances at early training stages, we formulate the RC loss in the *log domain*. Let $z_i = \log \Delta_i$, the loss is defined as the variance of log-IBIs:

$$\mathcal{L}_{\text{RC}} = \frac{1}{N-1} \sum_{i=1}^{N-1} (z_i - \bar{z})^2, \quad (5)$$

where $\bar{z} = \frac{1}{N-1} \sum_{i=1}^{N-1} z_i$ is the mean log-interval. The RC loss operates at the sequence level to encourage rhythmically consistent beat intervals, while the BCE loss provides frame-level supervision to ensure local prediction accuracy. To integrate these complementary objectives, we define the total training loss as a weighted combination:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda \cdot \mathcal{L}_{\text{RC}}, \quad (6)$$

where λ is a hyperparameter controlling the strength of rhythmic regularization.

2.4. Implementation Details

Our training pipeline processes 30-second raw audio waveforms ($T=30$) at 8192 Hz sampling rate ($S=8192$). Through three-stage downsampling in temporal compression encoder, we achieve a cumulative compression factor of 64, resulting in feature sequences at 128 Hz frame rate. This 7.8 ms resolution provides $2.6\times$ higher temporal acuity than standard beat annotation tolerances (± 20 ms), while reducing sequence length from 245760 to 3840, enabling efficient long-context modeling in subsequent Mamba blocks. All intermediate layers maintain a uniform channel dimension ($C=96$). For the rhythmic consistency loss, we set the RC weight to $\lambda=0.3$ with a short warm-up over the first 20% of training steps.

3. EXPERIMENTS

3.1. Datasets and Metrics

BeatMamba is trained and evaluated using seven widely adopted music datasets. For model training, we rely on three beat-annotated datasets: *Beatles* [18], *RWC Popular* [19], and *Harmonix* [20]. Evaluation is carried out on *Ballroom* [21, 22], *Hainsworth* [23], and *SMC* [24] through an 8-fold cross-validation scheme. *GTZAN* [25] is reserved as a completely independent test set. In addition, we adopt the augmentation strategies introduced in [14] to effectively enlarge the diversity of the training samples. For performance assessment on beat tracking tasks, we employ three standard metrics: the F1-measure, CMLt (Correct Metric Level), and AMLt (Allowed Metrical Level).

3.2. Results and Analysis

To comprehensively evaluate the performance of BeatMamba, we compare it against a series of representative state-of-the-art methods. These baselines span different architectural paradigms and reflect the evolution of beat tracking models, including the dilated convolutional TCN [11], the self-attention-driven Beat Transformer [13] and Beat This [14], and the kernel-attention convolution BeatKAN [26].

As shown in Table 1, BeatMamba achieves superior performance compared to previous state-of-the-art models across most datasets. On *Ballroom* and *Hainsworth*, BeatMamba attains the best CMLt and AMLt scores, indicating that its predictions are not only frame-level accurate but also structurally coherent in rhythm. On the cross-genre *GTZAN* dataset, BeatMamba consistently outperforms Transformer- and KAN-based models, demonstrating strong generalization across diverse musical styles. Although on the more challenging *SMC* dataset BeatMamba is slightly outperformed by BeatKAN [26], its performance remains competitive, highlighting the robustness of our method under complex rhythmic scenarios. Overall, the consistent improvements across multiple metrics confirm the effectiveness of BeatMamba.

Further comparisons show that incorporating the RC loss yields significant benefits on most datasets. On *Ballroom*, *Hainsworth* and *GTZAN*, adding RC loss clearly improves CMLt and AMLt, indicating that the predicted beats become more regular and consistent. Notably, we observe that RC loss results in negative gains on the *SMC* dataset. We attribute this to the fact that *SMC* mainly consists of Western classical music, which is characterized by complex rhythmic structures and frequent tempo fluctuations. Under such conditions, the strict isochronous constraint imposed by RC loss may hinder the model's ability to adapt to genuine tempo variations, leading to degraded performance. As future work, we plan to extend the RC loss with tempo-aware regularization, enabling it to preserve rhythmic consistency while better accommodating dynamic tempo changes in complex musical pieces.

Table 1. Comparison of beat tracking performance between BeatMamba and several state-of-the-art methods on four benchmark datasets.

Model	F-Measure	CMLt	AMLt
Ballroom			
TCN [11]	96.2	94.7	96.1
Beat Transformer [13]	96.8	95.4	96.6
BeatKAN [26]	96.7	93.6	96.3
BeatMamba w/o RCL	96.3	93.2	95.8
BeatMamba	96.5	95.7	97.2
Hainsworth			
TCN [11]	90.2	84.8	93.0
Beat Transformer [13]	90.2	84.2	91.8
BeatKAN [26]	91.3	84.6	94.6
BeatMamba w/o RCL	91.1	83.7	93.5
BeatMamba	91.5	85.2	95.3
SMC			
TCN [11]	54.4	44.3	63.5
Beat Transformer [13]	59.6	45.6	63.5
BeatKAN [26]	59.8	48.4	64.0
BeatMamba w/o RCL	59.0	47.3	62.5
BeatMamba	58.7	46.8	62.4
GTZAN			
TCN [11]	88.5	81.3	93.1
Beat Transformer [13]	88.5	80.0	92.2
Beat this [14]	88.9	79.9	89.4
BeatKAN [26]	88.2	78.1	92.3
BeatMamba w/o RCL	88.7	81.3	93.2
BeatMamba	88.9	82.3	94.3

3.3. Ablation Study

To assess the effectiveness of key design choices, we conduct ablation experiments focusing on two aspects: the directionality of the Mamba block and the weight of the RC loss.

Effect of Bidirectional Mamba Blocks. We compare the proposed bidirectional Mamba block against a unidirectional variant that only processes temporal sequences in the forward direction. As shown in Table 2, the bidirectional design provides consistent improvements across all metrics. This confirms that leveraging both past and future temporal contexts is crucial for robust rhythmic modeling.

Effect of RC Loss Weight. As shown in Fig. 3, we conduct a grid search over the weight $\lambda \in \{0.0, 0.1, \dots, 1.0\}$ with a step size of 0.1 on GTZAN dataset. Quantitative results demonstrate that setting $\lambda = 0.3$ for the RC loss yields significant and consistent improvements across all evaluation metrics relative to the baseline $\lambda = 0$. These findings indicate that a moderate rhythmic consistency regularization effectively suppresses irregular inter-beat spacing and enhances global metrical coherence without sacrificing frame-level ac-

Table 2. Comparison between unidirectional and bidirectional Mamba blocks on GTZAN dataset.

Architecture	F-Measure	CMLt	AMLt
Unidirectional Mamba	86.7	81.4	92.5
Bidirectional Mamba	88.9	82.3	94.3

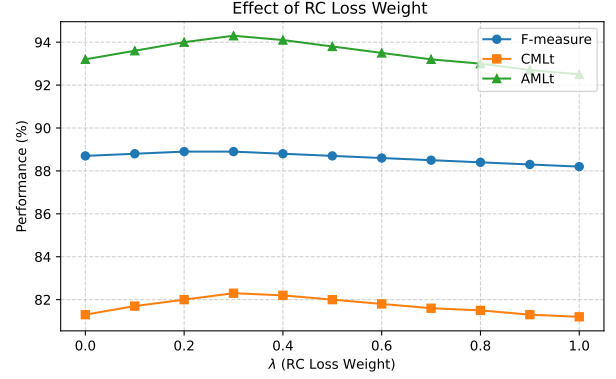


Fig. 3. Effect of different RC loss weights (λ) on beat tracking performance. The x-axis denotes λ values, and the y-axis shows performance metrics.

curacy. When $\lambda > 0.3$, all metrics start to decline. This decline stems from a critical trade-off: overly strong structural constraints impair the model’s sensitivity to crucial local acoustic cues, such as weak onsets and syncopations. The diminished capacity to detect these fine-grained events directly reduces the frame-level F1-measure, which in turn cascades to negatively impact sequence-level metrics including CMLt and AMLt.

4. CONCLUSION

We introduced BeatMamba, a beat-tracking framework that fuses local convolutional feature extraction with selective state-space modeling for global context. The architecture employs a symmetric U-shape built from Mamba blocks and bidirectional temporal scanning, enabling linear-time long-range modeling, while the proposed rhythmic consistency loss regularizes inter-beat intervals at the sequence level. Extensive evaluations on benchmark datasets confirm that BeatMamba delivers superior beat-tracking accuracy and metrical coherence, underscoring its efficiency and robustness on complex musical scenarios.

5. ACKNOWLEDGEMENT

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6. REFERENCES

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